

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

***CO3:** Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4
Marks:40

Total

Annexure A

CASE STUDY

Code of Conduct:

*I **JEROLIN MARY R (192411024)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- ii. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- iii. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: SoilDry $\rightarrow$ IrrigationNeeded
- 2) P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\neg$ Wumpus [x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Case Study 1 – Smart Medical Diagnosis System

(a) Representation using Propositional Logic

Step 1: Define the Symbols

F = Patient A has Fever

C = Patient A has Cough

R = Patient A has Rash

B = Patient A has Breathlessness

Fl = Patient A has Possible Flu

M = Patient A has Possible Measles

P = Patient A has Possible Pneumonia

Step 2: Represent the Rules

Rule I:

$$(F \wedge C) \rightarrow Fl$$

Rule II:

$$(F \wedge R) \rightarrow M$$

Rule III:

$$(C \wedge B) \rightarrow P$$

Step 3: Represent the Facts

F

C

B

Knowledge Base (KB)

Rules:

$$(F \wedge C) \rightarrow Fl$$

$$(F \wedge R) \rightarrow M$$

$$(C \wedge B) \rightarrow P$$

Facts:

F, C, B

(b) Apply Forward Chaining

Forward Chaining begins with the known facts and repeatedly applies the rules to derive new facts.

Initial Facts

{F, C, B}

Step 1

Check Rule I:

$(F \wedge C) \rightarrow FI$

Known facts:

F = True

C = True

Since both conditions are true, Rule I fires.

New fact derived:

FI (Possible Flu)

Knowledge Base becomes:

{F, C, B, FI}

Step 2

Check Rule II:

$(F \wedge R) \rightarrow M$

Known facts:

F = True

R = False (or not given)

Since Rash is not known, Rule II cannot be applied.

No new fact is derived.

Step 3

Check Rule III:

$(C \wedge B) \rightarrow P$

Known facts:

C = True

B = True

Both conditions are true.

Rule III fires.

New fact derived:

P (Possible Pneumonia)

Knowledge Base becomes:

{F, C, B, Fl, P}

Step 4

Check all rules again.

No more new facts can be derived.

Forward Chaining stops.

Forward Chaining Table

Step	Facts Available	Rule Fired	New Fact Derived
Initial	F, C, B	—	—
1	F, C	Rule I	Fl (Possible Flu)
2	F, R	Rule II	Cannot fire
3	C, B	Rule III	P (Possible Pneumonia)
Final	F, C, B, Fl, P	No further rules	Stop

Final Diagnoses

Possible Flu

Possible Pneumonia

Possible Measles cannot be concluded because the fact "Rash" is not present.

(c) Apply Backward Chaining

Goal

Prove:

P (Possible Pneumonia)

Step 1

Look for a rule that concludes P.

Rule III:

$$(C \wedge B) \rightarrow P$$

To prove P, we must first prove:

C

B

Step 2

Check the facts.

C = True

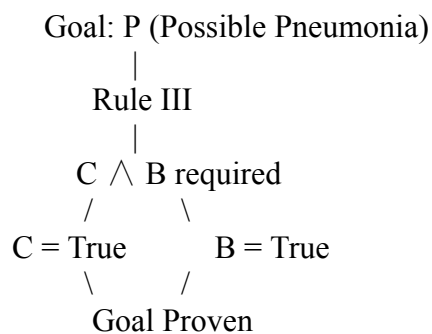
B = True

Since both required facts are already in the Knowledge Base, Rule III is satisfied.

Therefore,

P (Possible Pneumonia) is proved.

Goal Reduction Tree



Backward Chaining Table

Step	Current Goal	Rule Used	Sub-goals	Result
1	P	Rule III	C, B	Continue
2	C	Fact	—	True
3	B	Fact	—	True
4	P	Rule III satisfied	—	Goal Proven

Conclusion

The Smart Medical Diagnosis System uses propositional logic to represent medical knowledge and infer possible diseases.

Using Forward Chaining, the system derives two diagnoses:

- Possible Flu
- Possible Pneumonia

Using Backward Chaining, the goal "Possible Pneumonia" is successfully proved because both required facts (Cough and Breathlessness) are already present in the Knowledge Base.

Therefore, the system concludes that **Patient A has Possible Pneumonia and Possible Flu**, while **Possible Measles cannot be inferred due to the absence of the Rash symptom**.

Case Study 2 – Autonomous Traffic Management Agent

(a) Apply Unification

Given Rule 1

$\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Given Facts

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

$\text{Vehicle}(\text{CarA})$

$\text{Behind}(\text{CarA}, \text{Ambulance})$

Step 1: Match Predicate $\text{Vehicle}(x)$

Rule:

$\text{Vehicle}(x)$

Fact:

$\text{Vehicle}(\text{Ambulance})$

Substitution:

$\theta_1 = \{x/\text{Ambulance}\}$

The variable x is replaced by Ambulance.

Step 2: Match Predicate $\text{EmergencyType}(x)$

After applying θ_1 :

EmergencyType(Ambulance)

Fact:

EmergencyType(Ambulance)

The predicates match successfully.

Substitution remains:

$\theta_2 = \{x/\text{Ambulance}\}$

Step 3: Apply the Rule

Since both conditions are satisfied,

Vehicle(Ambulance)

EmergencyType(Ambulance)

Rule 1 fires.

Therefore,

ClearPath(Ambulance)

is derived.

Most General Unifier (MGU)

$\theta = \{x/\text{Ambulance}\}$

Unification Summary

Step	Predicate	Matching Fact	Substitution (θ)
1	Vehicle(x)	Vehicle(Ambulance)	$\{x/\text{Ambulance}\}$
2	EmergencyType(x)	EmergencyType(Ambulance)	$\{x/\text{Ambulance}\}$
3	Rule Applied	ClearPath(Ambulance)	Final MGU = $\{x/\text{Ambulance}\}$

(b) Resolution Method to Prove Proceed(CarA)

Step 1: Convert Rules into CNF

Rule 1:

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

CNF:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Rule 2:

$\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x)$

CNF:

$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

(We do not need the $\neg \text{RedSignal}$ part since it is not required for this proof.)

Rule 3:

$\text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

CNF:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Facts

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

$\text{Vehicle}(\text{CarA})$

$\text{Behind}(\text{CarA}, \text{Ambulance})$

Negated Goal

Goal:

$\text{Proceed}(\text{CarA})$

Negated Goal:

$\neg \text{Proceed}(\text{CarA})$

Resolution Steps

Resolution 1

Rule 1:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Using $\theta = \{x/\text{Ambulance}\}$

Resolve with

$\text{Vehicle}(\text{Ambulance})$

EmergencyType(Ambulance)

Result:

ClearPath(Ambulance)

Resolution 2

Rule 2:

$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

Using $x = \text{Ambulance}$

Resolve with

ClearPath(Ambulance)

Result:

GreenSignal(Ambulance)

Resolution 3

Rule 3:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Substitute

$x = \text{CarA}$

$y = \text{Ambulance}$

Rule becomes:

$\neg \text{Vehicle}(\text{CarA}) \vee \neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$

Resolve using facts:

Vehicle(CarA)

Behind(CarA, Ambulance)

GreenSignal(Ambulance)

Result:

Proceed(CarA)

Resolution 4

Resolve

Proceed(CarA)

with

$\neg$ Proceed(CarA)

Result:

□ (Empty Clause)

Since the empty clause is obtained, the theorem is proved.

Therefore,

Proceed(CarA) is TRUE.

Resolution Proof Tree

Vehicle(Ambulance)

EmergencyType(Ambulance)

|

Rule 1

|

ClearPath(Ambulance)

|

Rule 2

|

GreenSignal(Ambulance)

Vehicle(CarA)

Behind(CarA, Ambulance)

GreenSignal(Ambulance)

|

Rule 3

|

Proceed(CarA)

|

$\neg$ Proceed(CarA)

|

□

(c) Analysis of the Knowledge Base

The current knowledge base is sufficient for handling **one emergency vehicle**, but it is **not sufficient for handling two or more emergency vehicles simultaneously**.

Reason

The rules only describe how to clear the path for a single emergency vehicle. They do not specify what should happen when two emergency vehicles arrive at the same time.

For example,

EmergencyType(Ambulance1)

EmergencyType(FireTruck1)

Both vehicles may require a green signal simultaneously, which could create conflicting traffic directions. The current KB has no rule to resolve such conflicts.

Additional Rules Required

Rule 4: Priority Assignment

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{Priority}(x)$

This assigns a priority level to every emergency vehicle.

Rule 5: Conflict Resolution

$\text{Priority}(x) > \text{Priority}(y) \rightarrow \text{ClearPath}(x)$

This ensures that the higher-priority emergency vehicle receives the cleared route first.

Rule 6: Sequential Clearance

$\text{ClearPath}(x) \wedge \text{Waiting}(y) \rightarrow \text{ClearPath}(y)$

This allows the second emergency vehicle to receive a clear path after the first one has passed.

Additional Facts (Example)

Vehicle(FireTruck1)

EmergencyType(FireTruck1)

Priority(Ambulance1)

Priority(FireTruck1)

Justification

These additional rules enable the system to:

- Handle multiple emergency vehicles safely.
- Prevent conflicting green signals.
- Prioritize vehicles based on urgency.
- Ensure that all emergency vehicles eventually receive a clear route.

Without these rules, the knowledge base cannot make logical decisions when more than one emergency vehicle requires access at the same time.

Conclusion

Using unification, the system derives **ClearPath(Ambulance)** with the substitution set $\theta = \{x/\text{Ambulance}\}$. Using the resolution method, it successfully proves **Proceed(CarA)** by deriving **GreenSignal(Ambulance)** and obtaining the empty clause ($\square$). However, the current FOL knowledge base is designed for a single emergency vehicle. To support multiple simultaneous

emergency vehicles, additional rules for **priority management, conflict resolution, and sequential path clearance** are required.

Case Study 3 – Agricultural Expert Reasoning System

(a) Convert the Knowledge Base into Conjunctive Normal Form (CNF)

Define the Symbols

SD = SoilDry

IN = IrrigationNeeded

CW = CropWheat

ADM = ApplyDripMethod

CAR = CropAtRisk

P1: SoilDry $\rightarrow$ IrrigationNeeded

Step 1: Original Statement

$SD \rightarrow IN$

Step 2: Eliminate Implication

$\neg SD \vee IN$

Step 3: Move Negation Inward

No change required.

Step 4: Distribute

Already in CNF.

Final CNF

$\neg SD \vee IN$

P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

Step 1: Original Statement

$(IN \wedge CW) \rightarrow ADM$

Step 2: Eliminate Implication

$$\neg(\text{IN} \wedge \text{CW}) \vee \text{ADM}$$

Step 3: Move Negation Inward (De Morgan's Law)

$$(\neg\text{IN} \vee \neg\text{CW}) \vee \text{ADM}$$

Step 4: Distribute

Already in CNF.

Final CNF

$$\neg\text{IN} \vee \neg\text{CW} \vee \text{ADM}$$

P3: $\neg\text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Step 1: Original Statement

$$(\neg\text{ADM} \wedge \text{CW}) \rightarrow \text{CAR}$$

Step 2: Eliminate Implication

$$\neg(\neg\text{ADM} \wedge \text{CW}) \vee \text{CAR}$$

Step 3: Move Negation Inward

$$\text{ADM} \vee \neg\text{CW} \vee \text{CAR}$$

Step 4: Distribute

Already in CNF.

Final CNF

$$\text{ADM} \vee \neg\text{CW} \vee \text{CAR}$$

Facts

SoilDry = True

CropWheat = True

CNF:

SD

CW

Final CNF Knowledge Base

1. $\neg\text{SD} \vee \text{IN}$
2. $\neg\text{IN} \vee \neg\text{CW} \vee \text{ADM}$
3. $\text{ADM} \vee \neg\text{CW} \vee \text{CAR}$

4. SD
5. CW

(b) Resolution Refutation to Prove ApplyDripMethod

Goal

Prove:

ADM

Step 1: Negate the Goal

$\neg$ ADM

Add this clause to the CNF knowledge base.

CNF Clauses

C1: $\neg$ SD $\vee$ IN

C2: $\neg$ IN $\vee$ $\neg$ CW $\vee$ ADM

C3: ADM $\vee$ $\neg$ CW $\vee$ CAR

C4: SD

C5: CW

C6: $\neg$ ADM

Resolution Steps

Step 1

Resolve C1 and C4.

C1: $\neg$ SD $\vee$ IN

C4: SD

Result:

IN

Step 2

Resolve C2 with IN.

C2: $\neg$ IN $\vee$ $\neg$ CW $\vee$ ADM

IN

Result:

$\neg CW \vee ADM$

Step 3

Resolve the above clause with CW.

$\neg CW \vee ADM$

CW

Result:

ADM

Step 4

Resolve ADM with the negated goal.

ADM

$\neg ADM$

Result:

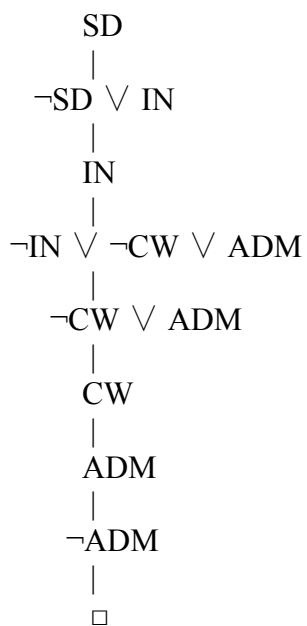
□ (Empty Clause)

Since the empty clause is obtained, the theorem is proved.

Therefore,

ApplyDripMethod is TRUE.

Resolution Proof Tree



(c) Effect of Removing the Fact "SoilDry"

Suppose the fact

SD (SoilDry)

is removed from the knowledge base.

The remaining facts are:

CW

Step 1

Rule P1:

$SD \rightarrow IN$

Since SD is no longer available, we cannot derive:

IN

Step 2

Rule P2:

$IN \wedge CW \rightarrow ADM$

We know:

$CW = \text{True}$

But

$IN = \text{Unknown}$

Since both conditions are not satisfied, ApplyDripMethod cannot be concluded.

Step 3

Rule P3:

$\neg ADM \wedge CW \rightarrow CAR$

Although CW is true, we have **not proved** $\neg ADM$.

Failing to prove ADM is **not the same as proving** $\neg ADM$.

Therefore, the condition $\neg ADM$ is not satisfied.

Hence, CropAtRisk cannot be concluded.

Reasoning

Without the fact SoilDry:

- IrrigationNeeded cannot be inferred.
- ApplyDripMethod cannot be inferred.
- $\neg$ ApplyDripMethod is also not known to be true.
- Therefore, CropAtRisk does not logically follow.

The reasoning fails because one of the required premises is missing.

Conclusion

When SoilDry is removed from the knowledge base, the inference chain is broken. The system cannot derive IrrigationNeeded, which means ApplyDripMethod cannot be proved. Since $\neg$ ApplyDripMethod is also not established, the rule for CropAtRisk cannot be applied. Therefore, **CropAtRisk does not logically follow after removing the SoilDry fact.**

Case Study 4 – Wumpus World Knowledge Agent

(a) Construct the Agent's Belief State and Apply Modus Ponens

Given Percepts

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]

Knowledge Base Rules

Rule 1:

$\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell } [1,2]$

Rule 2:

$\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell } [1,1]$

Rule 3:

$\text{Glitter}[x,y] \rightarrow \text{Gold is at } [x,y]$

Rule 4:

$\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

Agent's Initial Belief State

Known Facts:

$\text{Stench}[1,2]$

Breeze[1,1]

Glitter[2,2]

Apply Modus Ponens

Step 1

Fact:

Stench[1,2]

Rule:

$\text{Stench}[1,2] \rightarrow \text{Wumpus adjacent to } [1,2]$

Derived Fact:

Wumpus is adjacent to cell [1,2]

Step 2

Fact:

Breeze[1,1]

Rule:

$\text{Breeze}[1,1] \rightarrow \text{Pit adjacent to } [1,1]$

Derived Fact:

Pit is adjacent to cell [1,1]

Step 3

Fact:

Glitter[2,2]

Rule:

$\text{Glitter}[x,y] \rightarrow \text{Gold}[x,y]$

Substitute:

$x = 2$

$y = 2$

Derived Fact:

Gold is at [2,2]

Step 4

Rule 4 requires knowledge that a cell has neither a Wumpus nor a Pit.

Since such information is not available, no Safe cell can be inferred.

Derived Conclusions

- Wumpus is adjacent to cell [1,2].
- Pit is adjacent to cell [1,1].
- Gold is located at cell [2,2].
- No cell can be confirmed as safe.

(b) Apply Forward Chaining

Forward Chaining begins with the known percepts and repeatedly applies the rules.

Initial Facts

Stench[1,2]

Breeze[1,1]

Glitter[2,2]

Step 1

Apply Rule 1.

Stench[1,2]

↓

Wumpus is adjacent to [1,2]

Cells adjacent to [1,2] are:

[1,1]

[1,3]

[2,2]

Therefore, the possible Wumpus locations are:

[1,1]

[1,3]

[2,2]

Step 2

Apply Rule 2.

Breeze[1,1]

↓

Pit is adjacent to [1,1]

Cells adjacent to [1,1] are:

[1,2]

[2,1]

Therefore, the possible Pit locations are:

[1,2]

[2,1]

Step 3

Apply Rule 3.

Glitter[2,2]

↓

Gold is at [2,2]

Forward Chaining Summary

Step	Rule Applied	New Fact
1	Rule 1	Wumpus adjacent to [1,2]
2	Rule 2	Pit adjacent to [1,1]
3	Rule 3	Gold at [2,2]

Possible Locations

Possible Wumpus:

- [1,1]
- [1,3]
- [2,2]

Possible Pit:

- [1,2]
- [2,1]

Gold:

- [2,2]

(c) Evaluate the Path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$

The proposed path is:

$[1,1] \rightarrow [1,2] \rightarrow [2,2]$

Step 1

At cell $[1,1]$, the agent perceives a Breeze.

This indicates that a Pit exists in one of the adjacent cells.

Possible Pit locations:

- $[1,2]$
- $[2,1]$

Since $[1,2]$ is on the planned path, it may contain a Pit.

Step 2

At cell $[1,2]$, the agent perceives a Stench.

This indicates that the Wumpus is in one of the adjacent cells.

Possible Wumpus locations:

- $[1,1]$
- $[1,3]$
- $[2,2]$

Since $[2,2]$ is on the planned path, it may contain the Wumpus.

Step 3

The agent knows that Gold is located at $[2,2]$.

However, Rule 4 states that a cell is safe only if it is known to contain neither a Wumpus nor a Pit.

The agent cannot prove:

$\neg \text{Wumpus}[2,2]$

or

$\neg \text{Pit}[1,2]$

Therefore, the path cannot be confirmed as safe.

Conclusion

The path:

$[1,1] \rightarrow [1,2] \rightarrow [2,2]$

is **not guaranteed to be safe** because:

- Cell [1,2] may contain a Pit.
- Cell [2,2] may contain the Wumpus.
- Although the Gold is at [2,2], the agent does not have enough information to prove that these cells are free of danger.

Therefore, the logical agent should **not follow this path until additional percepts or knowledge confirm that the cells are safe.**
